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Chapter 8

Interest parity

International finance is the art of borrowing on the strength of what you already owe.

Evan Esar

The purchasing-power-parity (PPP) condition considered in Chapter 8, applies to product markets. There is another important, parallel parity condition that applies to financial markets. This is the **covered interest-parity condition**. It states that when steps have been taken to avoid foreign exchange risk by use of forward contracts, rates of return on investments, and costs of borrowing, will be equal irrespective of the currency of denomination of the investment or the currency borrowed.

In this chapter we derive the covered interest-parity condition and show its connection to the PPP principle. We also consider the "frictions" that must be absent for the covered interest-parity condition to hold. The frictions that must be absent include legal restrictions on the movement of capital, transaction costs, and taxes. These frictions play an analogous role to the frictions that must be absent for PPP to hold, namely, restrictions on the movement of goods between markets, transportation costs, and tariffs.

Our approach to deriving the covered interest-parity condition begins by explaining how to make short-term investment and borrowing decisions in the international context. We then show how shopping around for the highest investment yield or lowest borrowing cost pushes yields and costs in different currencies towards equality, thereby resulting in covered interest parity. Our focus is on investment yields and borrowing costs in different *currencies*, and not different *countries*, because as we shall see, securities are often denominated in different currencies within a single country. For most purposes in this book the *currency* of denomination is more important than the *country* in which a security is issued. Currency of denomination introduces foreign exchange risk while country of issue introduces political risk, and for most countries and time periods, foreign exchange risk is a far larger concern than political risk.

As we proceed in this and later chapters, it will be useful at a number of points to develop and illustrate concepts by referring to an example. For this purpose, we shall consider a manufacturing company that makes denim clothing, primarily jeans. The company is called Aviva Corporation. Aviva is headquartered in the United States but has sales, and purchases its denim, in many different countries. For this chapter, the important characteristic about Aviva is that it has uneven cash flows; therefore, on some occasions it has surplus funds to invest, and on other occasions it needs to borrow.

Short-term borrowing and investment take place in the **money market**. This is the market in which short-term securities such as treasury bills and commercial paper are traded. Because

there are actively traded forward contracts with relatively short-term money-market maturities, the money market deserves special treatment. Forward contracts allow money-market borrowers and investors to avoid foreign exchange risk and exposure. Foreign exchange risk and exposure are discussed in some detail in Chapter 9. For the time being we note that they are the result of sensitivity and uncertainty in asset or liability values due to unexpected changes in exchange rates.[1] Let us start by asking in which currency Aviva should invest.

THE INVESTMENT AND BORROWING CRITERIA

Determining the currency of investment

Suppose that a firm like Aviva Corporation has some funds to place in the money market for 3 months. Perhaps it has received a major payment but can wait before paying for a large investment in new equipment. The firm could place these funds in securities denominated in its own domestic currency at an interest rate that can be discovered simply by calling around for the going rates on, for example, locally traded commercial paper or treasury bills. Alternatively, it could invest in foreign currency-denominated securities. Should it buy money market securities denominated in domestic or in foreign currency?

Many countries have money markets in which financial securities denominated in the countries' own currencies are actively traded. For example, there is a well-developed market in Canadian dollar securities in Canada, of pound securities in Britain, and of yen securities in Japan. Furthermore, in large international financial centers such as London and New York, there are active markets in securities denominated in a variety of different foreign currencies. For example, in London there are active markets in US dollar securities, euro securities, yen securities, Swiss franc securities and other currency-denominated instruments, as well, of course, as securities denominated in British pounds. By glancing at quoted interest rates it might sometimes seem possible to obtain higher yields on some foreign currency-denominated securities than on others. However, realized yields on foreign-currency securities depend on what happens to exchange rates as well as on interest rates. If, for example, the value of the foreign currency in which Aviva's investments are denominated happens to fall unexpectedly before maturity, then there will be a foreign exchange loss when the investments are converted back into dollars. As we shall see, the existence of the forward exchange market allows us to compute yields which include the effects of exchange rates. However, we shall also see that when the forward market is used to remove foreign exchange risk, yield differences on different currency-denominated securities are likely to be very small.

Let us see how an exchange-risk-free investment decision is made. We will select for our example Aviva's choice between alternative 3-month rather than full-year securities to make clear the need to keep exchange-rate movements and interest rates in comparable annualized terms.

Aviva knows that if it puts its funds in a US-dollar investment such as a bank deposit for 3 months, at maturity each dollar will provide

$$\$\left(1+\frac{r_\$}{4}\right)$$

where $r_\$$ is the annualized US dollar interest rate, and division by 4 gives the 3-month return.[2] The interest rate is measured in decimal form, so a 5-percent rate is expressed as $r_\$ = 0.05$.

1 Risk can exist even if values of assets and liabilities, are not uncertain. Risk also exists if there is uncertainty in the prices of what people buy, so-called **inflation risk**, due to uncertainty in the buying power of given amounts of money.

2 Later, we allow for compound interest in computing returns. However, division by 4 is expositionally more convenient than taking the fourth root to find the 3-month return.

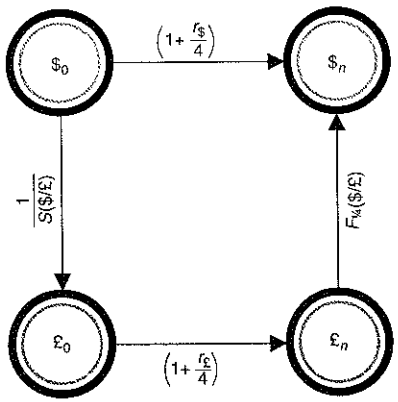


Figure 8.1 *Dollar versus hedged pound investments*

Notes
A US investor with funds to invest for 3 months can simply invest in domestic-currency denominated 3-month securities. After 3 months, the payout per dollar invested is as shown on the top horizontal arrow pointing from time zero to maturity, one-quarter of a year later. Alternatively, the investor can buy pounds spot, shown by the left, downward-pointing arrow, and invest in pound-denominated securities, shown by the lower rightward-pointing arrow. The pound investment can be hedged by selling the pounds forward for dollars, shown by the right upward-pointing arrow. The payout from hedged pound investment is given by the multiplication of the relevant amounts on the three sides of the figure, left, bottom, and right.

This alternative is illustrated in Figure 8.1 with the horizontal arrow from $\$_0$ to $\$_n$. The rightward pointing arrow indicates an investment, moving money from the current period, time zero, to a later period, n, in this case a quarter of a year.

Suppose that Aviva considers investing in a pound-denominated bank deposit, and that the spot dollar/sterling exchange rate, in the conventional US terms of quotation, is $S(\$/£)$. The exchange rate $S(\$/£)$ gives the number of dollars per pound sterling, and so for \$1 Aviva will obtain $1/S(\$/£)$ in British pounds, assuming that there are no transaction costs. This is illustrated in Figure 8.1 with the downward-pointing arrow from $\$_0$ to $£_0$.

If the annualized interest rate on 3-month British pound bank deposits is $r_£$, then for every dollar invested, Aviva will receive after 3 months the number of pounds that was invested (the principal), $1/S(\$/£)$, plus the 3-month interest on this, which is the principal multiplied by $r_£/4$. This is illustrated in Figure 8.1 with the rightward-pointing arrow from $£_0$ to $£_n$ where in this case $n = \frac{1}{4}$. That is, Aviva will receive

$$£\frac{1}{S(\$/£)}\left(1+\frac{r_£}{4}\right) \tag{8.1}$$

For example, if $S(\$/£) = 1.5000$ and $r_£ = 0.06$, then each dollar invested in pound-denominated bank deposits will provide after 3 months:[3]

$$£\frac{1}{1.5000}\left(1+\frac{0.06}{4}\right) = £0.6767$$

This certain number of pounds represents an uncertain number of dollars, but a forward contract can offer a complete hedge and guarantee the number of dollars that will be received for the pounds at the maturity of the security.

If, at the time of buying the 3-month pound-denominated deposit, Aviva sells forward the amount of pounds to be received at maturity, that is, the amount in equation (8.1) or £0.6767 in our example, then the number of dollars that will be obtained is set by the forward contract. After 3 months, Aviva delivers the British pounds and receives the number of dollars stated in the forward contract. This is illustrated in Figure 8.1 with the upward-pointing arrow from $£_n$ to $\$_n$.

If, for example, the 3-month forward rate at the time of investment is $F_{1/4}(\$/£)$, then we multiply the amount in equation (8.1) by this exchange rate to find the number of dollars received at maturity for each original dollar invested in the pound-denominated bank deposit. We obtain

$$\$\frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right) \tag{8.2}$$

For example, if $F_{1/4}(\$/£) = 1.4950$, then the number of pounds from equation (8.1), approximately £0.6767, will provide (\$1.4950/£) × £0.6767 = \$1.0116 when sold forward for dollars.

3 We round to the fourth decimal.

This is the number of dollars received after 3 months, or $\frac{1}{4}$ of a year, for each original dollar in the pound-denominated bank deposit. This implies an annual rate of return of approximately,

$$4\left(\frac{1.0116 - 1.0000}{1.0000}\right) = 0.04647, \text{ or } 4.65\%$$

It is important to remember that the number of dollars given in equation (8.2) is a certain amount that is known at the time of investment. The purchase of the spot pounds, the investment in the pound-denominated deposit, and the forward sale of pounds all take place at the same time, and so if the security itself is risk-free there is no doubt about the number of dollars that will be received. If the spot exchange rate changes before the deposit matures, that will make no difference. The exchange rate to be used in converting the dollars into pounds at maturity is already set in the forward contract, which is part of the swap of dollars for pounds (recall from Chapter 3 that a swap is any exchange of currencies that is reversed, in this case dollars into pounds in the spot market and back to dollars in the forward market). In terms of our example, it is guaranteed that $1.0116 will be received.

It is now a simple matter to express the rule for deciding the currency in which to invest. The investor should choose a 3-month US-dollar deposit, rather than a pound deposit whenever[4]

$$\left(1+\frac{r_\$}{4}\right) > \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right)$$

The investor should select the pound deposit rather than the US-dollar deposit whenever the reverse inequality holds, that is

$$\left(1+\frac{r_\$}{4}\right) < \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right)$$

Only if

$$\left(1+\frac{r_\$}{4}\right) = \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right) \qquad (8.3)$$

should the investor be indifferent, since the same amount will be received from a dollar invested in securities denominated in either currency.[5]

We can convert equation (8.3) into a more meaningful equality if we subtract $(1 + r_£ / 4)$ from both sides:

$$\left(1+\frac{r_\$}{4}\right) - \left(1+\frac{r_£}{4}\right) = \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right) - \left(1+\frac{r_£}{4}\right)$$

With cancellation and rearrangement we obtain

$$r_\$ = r_£ + 4\left[\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}\right] \times \left(1+\frac{r_£}{4}\right) \qquad (8.4)$$

We interpret this equation below, but before we do, we can note that part of the second right-hand term in equation (8.4) involves the multiplication of two small numbers, the forward pound premium and $r_£/4$. This product is very small. For example, if the forward premium is 4 percent and British interest rates are 8 percent per annum, the cross-product term from equation (8.4) will be 0.0008 (0.04×0.02), which is less than one-tenth of 1 percent. In order to interpret equation (8.4), we might therefore temporarily drop the term formed from this product (which means dropping the $r_£/4$) and write it as:

$$r_\$ = r_£ + 4\left[\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}\right] \qquad (8.5)$$

4 When exchange rates are in European terms, the forward and spot rates must be inverted. This is left as an exercise at the end of the chapter.

5 In more general terms, equation (8.3) can be written as

$$\frac{F_n}{S} = \frac{(1+r)^n}{(1+r_f)^n}$$

where the form of the exchange-rate quotation and the annualization are assumed to be understood. Here, r is the domestic interest rate and r_f is the foreign currency interest rate.

The first term on the right of equation (8.5) is the annualized pound interest rate. The second right-hand term is the annualized (because of the 4) forward premium/discount on pounds. Therefore, we can interpret equation (8.5) as saying that investors should be indifferent between home- and foreign-currency denominated securities if the home-currency interest rate equals the foreign-currency rate plus the annualized forward exchange premium/discount on the foreign currency. Investors should invest in the home currency when the domestic-currency interest rate exceeds the sum of the foreign-currency rate plus the foreign exchange premium/discount, and invest abroad when the domestic-currency rate is less than this sum. We discover that a mere comparison of interest rates is not sufficient for making investment choices. In order to determine in which currency-denominated securities to invest, we must add the foreign-currency interest rate to the forward premium or discount. Using the terminology of Chapter 3, we must add the foreign-currency interest rate to the spot-forward swap of pounds for dollars, where this swap is put in annualized terms.

The difference between equation (8.4) and equation (8.5) is that in equation (8.4) we include the forward exchange premium/discount on the principal invested in pound securities *and* the forward exchange premium/discount on the interest earned. In the approximate form equation (8.5) we consider the forward premium/discount earned on the principal, but not the premium/discount on the interest.

An example: comparing investments

Suppose Aviva Corporation faces the exchange rate and interest rate situation shown in Table 8.1 and has $10 million to invest for 3 months. Into which currency-denominated bank deposit should it place its funds?

The yield on pound-denominated deposits when the proceeds are sold forward, called the **covered** or **hedged yield**, can be calculated from

$$r_£ + 4\left[\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}\right]\left(1+\frac{r_£}{4}\right)$$

where the first element is the pound interest rate and the second element is the premium/discount on the pound vis-à-vis the dollar, including the premium/discount on both the principal and interest. Substituting the values in Table 8.1

$$\text{British pound covered yield} = 0.05125 + 4\left(\frac{1.5121 - 1.5140}{1.5140}\right) \times \left(1+\frac{0.05125}{4}\right) = 0.046167, \text{ or } 4.6167\%$$

This yield, which involves no foreign exchange risk because the pounds are sold forward, is slightly higher than the yield on US-dollar deposits.

Similarly, the covered yields on the euro is

$$\text{Euro covered yield} = 0.07430 + 4\left(\frac{1.1175 - 1.1258}{1.1258}\right) \times \left(1+\frac{0.07430}{4}\right) = 0.044262, \text{ or } 4.4262\%$$

Table 8.1 *Exchange rates and interest rates on different currency-denominated 3-month bank deposits*

	US dollar	British pound	Euro	Swiss franc	Japanese yen
Interest rate[a]	4.4375%	5.1250%	7.4300%	3.9375%	1.125%
Spot rate (US equivalent)	1.0	$1.5140/£	$1.1258/€	SFr 1.4065/$ ($0.710985/SFr)	¥114.12/$ ($0.0087627/¥)
Forward rate[b] (US equivalent)	1.0	$1.5121/$	$1.1175/€	SFr 1.4052/$ ($0.711642/SFr)	¥113.20/$ ($0.0088339/¥)
Covered yield	4.4375%	4.6167%	4.4262%	4.3112%	4.3843%

Notes
a Interest rates on 3-month time deposits.
b 3-month forward rate.

For the covered yield on the Swiss franc and Japanese yen hedged against the US dollar we use the exchange rates in US-dollar terms shown in parentheses below the European-terms quotations. For example, for the covered yield on Swiss franc deposits we compute

Swiss franc covered yield

$$= 0.039375 + 4\left(\frac{0.711642 - 0.710985}{0.710985}\right) \times \left(1 + \frac{1.039375}{4}\right)$$

$$= 0.043108, \text{ or } 4.3108\%$$

The *covered* yields in Table 8.1 are much closer in value than the yields in the local currencies. An inspection of the spot versus forward exchange rates shows why. The British pound and euro which have higher interest rates than are available on US dollar securities, are both at a forward discount. The discounts offset the higher foreign currency interest rates. On the other hand, the Swiss franc and Japanese yen, the currencies with lower interest rates than on the US-dollar deposits, are both at a forward premium. The premiums help make up for the lower interest rates.

If Aviva were to invest its $10 million for 3 months in the British pound deposits, covered in the forward market, it would receive back,

$$\$10{,}000{,}000\left(1 + \frac{0.046167}{4}\right) = \$10{,}115{,}415$$

However, if Aviva had chosen US-dollar deposits it would have received back:

$$\$10{,}000{,}000\left(1 + \frac{0.044375}{4}\right) = \$10{,}109{,}375$$

The difference between the two paybacks is $6,040. This is the reward for Aviva doing its homework and finding the covered yield on different currency-denominated investments.

Determining the currency in which to borrow

Imagine that Aviva Corporation needs to borrow for 3 months. If the annualized interest rate for domestic-currency borrowing is $r_\$$, then the required repayment after 3 months is the principal plus interest, or

$$\$\left(1 + \frac{r_\$}{4}\right) \tag{8.6}$$

for each dollar borrowed. This is illustrated by the upper leftward-pointing horizontal arrow in Figure 8.2, showing borrowing as bringing funds to the current time period.

Suppose that instead of borrowing dollars directly, Aviva considers obtaining dollars indirectly by using a pound-denominated loan, and converting the borrowed pounds into the dollars it needs. If the spot exchange rate is $S(\$/£)$, then borrowing $1 requires borrowing $1/S(\$/£)$ in pounds. This is illustrated by the left-hand, upward-pointing arrow in Figure 8.2. For example, at the exchange rate $S(\$/£) = 1.5140$, borrowing

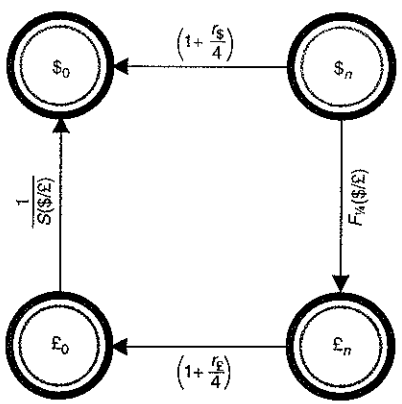


Figure 8.2 *Dollar versus hedged pound borrowing*

Notes
A US borrower seeking funds for 3 months can simply borrow dollars. The repayment at the end of the 3 months is shown on the top horizontal arrow pointing leftwards from maturity of the loan to time zero. Alternatively, the borrower can borrow pounds – the bottom leftward-pointing arrow – and use these to buy dollars spot – the left upward-pointing arrow. In this way the borrower has use of dollars. The pound loan can be hedged by buying the pounds forward, shown by the right downward-pointing arrow. The repayment for hedged pound borrowing is given by the multiplication of the relevant amounts on the three sides of the figure, left, bottom, and right.

$1 means borrowing £0.6605. If the annualized interest rate is $r_£$, then for each dollar borrowed via pounds, Aviva must repay

$$£\frac{1}{S(\$/£)}\left(1 + \frac{r_£}{4}\right) \tag{8.7}$$

This is illustrated in Figure 8.2 by multiplication of the amount on the lower leftward-pointing horizontal arrow and the amount on the left-hand, upward-pointing arrow. For example, if $r_£ = 0.06$ (the 3-month *annualized* borrowing rate), Aviva must repay £0.6605 × 1.015 = £0.6704. Without a forward exchange contract the number of dollars this would represent when Aviva repays its pound debt is uncertain. However, with a forward exchange contract the risk is eliminated.

Suppose that Aviva buys forward the amount of pounds in equation (8.7) at $F_{1/4}(\$/£)$. When the debt is repaid, Aviva will receive the required number of pounds on the forward contract for which it must pay

$$\$\frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1 + \frac{r_£}{4}\right) \tag{8.8}$$

The forward hedging is illustrated by the right-hand downward-pointing arrow in Figure 8.2. For example, if $F_{1/4}(\$/£) = 1.5121$, repaying £0.6704 involves paying $1.0137 ($1.5121/£ × £0.6704). On the other hand, if Aviva borrowed dollars for 3 months at $r_\$ = 0.05$, or 5 percent per annum, it would have to repay $1.0125 on each dollar. In this case of dollar borrowing, it is cheaper to borrow dollars directly than incur a hedged loan in pounds. In general, a firm should borrow pounds via a swap whenever the amount in equation (8.8) is less than that in equation (8.6), that is, when

$$\frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1 + \frac{r_£}{4}\right) < \left(1 + \frac{r_\$}{4}\right)$$

A firm should borrow dollars when the reverse inequality holds. In our particular example Aviva should borrow in dollars: the amount to be repaid is lower from a direct dollar loan. The borrowing decision criterion is seen to be the same as the investment criterion with, of course, the inequality reversed.

Borrowing and investing for arbitrage profit

Imagine a firm that can borrow its own currency and/or a foreign currency, as can a large corporation or bank. Suppose that it can borrow dollars for 3 months at an annualized interest rate of $r_\$$. Thus, for each dollar it borrows, it must repay $(1 + r_\$/4)$ dollars. This is illustrated by the leftward-pointing upper arrow in Figure 8.3. The firm can take each borrowed dollar and buy $1/S(\$/£)$ pounds, illustrated by the left, downward-pointing arrow in Figure 8.3. If these pounds are invested for 3 months at $r_£$ per annum, and if the resulting receipts are sold forward, the firm will receive

$$\$\frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1 + \frac{r_£}{4}\right)$$

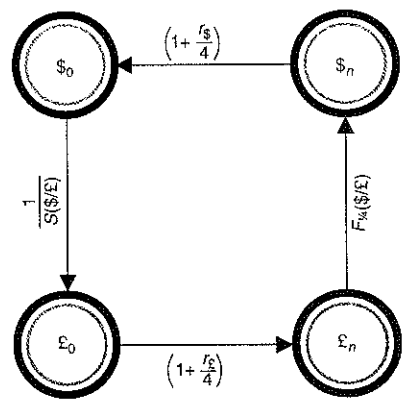


Figure 8.3 *Covered interest arbitrage: dollar borrowing and pound investing*

Notes
An interest arbitrager may begin by borrowing dollars – the upper leftward-pointing arrow. These borrowed dollars can be used to buy pounds spot – the left downward-pointing arrow. The pounds can then be invested for the same maturity as the dollar loan. With pounds maturing from the investment and dollars to be repaid on the loan there is exchange rate exposure and risk. The exposure and risk can be avoided by selling pounds forward for dollars – the right upward-pointing arrow. Exchange rates and interest rates adjust so that interest arbitrage opportunities do not persist.

The pound investment and forward sale of pounds are illustrated by the lower rightward-pointing and right-hand upper pointing arrows in Figure 8.3. Note that the company has begun with no funds of its own, and has taken no risk. Borrowing in dollars and simultaneously investing in pounds will result in a profit if the number of dollars received from the hedged pound investment exceeds the repayment on the dollar loan, that is, if

$$\left(1+\frac{r_\$}{4}\right) < \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right)$$

The reverse activity, borrowing in pounds and investing in dollars, will be profitable if the reverse inequality holds. As long as either inequality holds, it pays to borrow in one currency and lend, or invest, in the other. Borrowing and investing in this way with exchange rate risk hedged in the forward market is known as **covered interest arbitrage**. In terms of Figure 8.3, covered interest arbitrage borrowing of dollars and investing in pounds is seen to be a complete circuit around the figure in a counter-clockwise direction. The alternative covered interest arbitrage of borrowing pounds and investing in dollars would be a clockwise circuit around the figure.

It should be no surprise that the potential for covered interest arbitrage helps guarantee that little opportunity for profit remains, and that investors and borrowers will be relatively indifferent with regard to choosing a currency. This is clear, for example, from the similarity of covered yields in Table 8.1.

THE COVERED INTEREST-PARITY CONDITION

Mathematical statement of interest parity

We have determined that 3-month investors and borrowers would be indifferent between dollar and pound denominations of investment or debt if

$$\left(1+\frac{r_\$}{4}\right) = \frac{F_{1/4}(\$/£)}{S(\$/£)}\left(1+\frac{r_£}{4}\right) \qquad (8.9)$$

More precisely, if we allow for compound interest, as we should for long-term investing and borrowing, investors and borrowers will be indifferent between the dollar and pound for investing and borrowing when

$$(1+r_\$)^n = \frac{F_n(\$/£)}{S(\$/£)}(1+r_£)^n \qquad (8.10)$$

When equation (8.10) holds, no covered interest arbitrage is profitable. Equation (8.10) is the **covered interest-parity condition**. When this condition holds, there is no advantage to covered borrowing or investing in any particular currency, and no profit from any covered interest arbitrage.

The covered interest-parity condition is the financial-market equivalent of the law of one price from the commodity market, and follows from financial-market efficiency. The market forces leading to covered interest parity, as well as the factors which might result in small deviations from the parity condition, can be illustrated graphically.

Market forces resulting in covered interest-parity: a graphical presentation

We can represent covered interest parity by using the framework of Figure 8.4. The annualized 3-month forward premium on the pound – on principal plus interest – is drawn on the vertical axis, and the annualized interest rate difference between the dollar and the pound is drawn along the horizontal axis. The area above the horizontal axis represents a pound forward premium, and the area below this axis represents a pound forward discount. To the right of the vertical axis there is a dollar interest advantage, and to the left there is a dollar interest disadvantage.

Covered interest parity, as expressed in equation (8.4), can be written as

$$r_\$ - r_£ = 4\left[\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}\right] \times \left(1+\frac{r_£}{4}\right) \qquad (8.11)$$

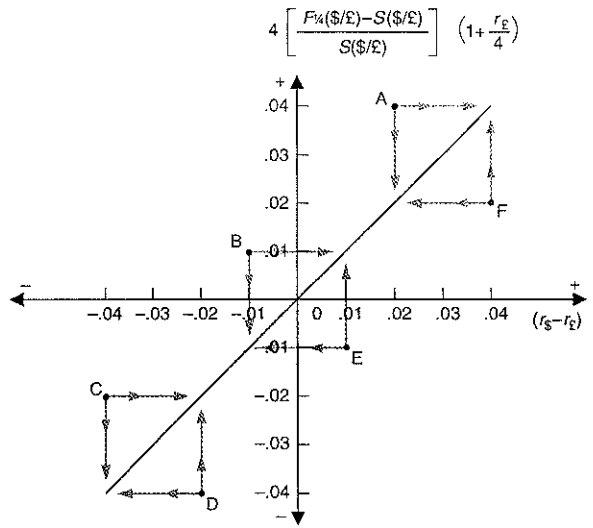


Figure 8.4 *The covered interest-parity diagram*

Notes
The diagonal is the line of covered interest parity. On the line, investors and borrowers are indifferent between dollar and pound investing/borrowing. Above and to the left of the line there is an incentive to invest in pounds and borrow dollars. Below and to the right of the line there is an incentive to invest in dollars and borrow pounds. In situations off the interest parity line, forces are at work pushing us back towards it.

If the same scale is used on the two axes in Figure 8.4, this parity condition is represented by a 45-degree line. This line traces the points where the two sides of our equation are indeed equal.

Suppose that instead of having equality as in equation (8.11), we have the following inequality:

$$r_\$ - r_£ < 4\left[\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}\right] \times \left(1+\frac{r_£}{4}\right) \qquad (8.12)$$

This condition means for example, that any pound forward premium more than compensates for any dollar interest advantage. Thus:

1 Covered investment in pounds yields more than in dollars.
2 Borrowing in dollars is cheaper than covered borrowing in pounds.

It also means that it is profitable for an interest arbitrager to borrow dollars and make a covered investment in pounds. Because this act of covered interest arbitrage involves borrowing in the cheaper currency, *and* investing in the higher yielding currency, we can concentrate on the consequences of interest arbitrage rather than the separate activities of borrowing and investing.

The incentive in equation (8.12) to borrow dollars and make a covered investment in pounds means an incentive to:

1 Borrow dollars, perhaps by issuing and selling a security – thus tending to increase $r_\$$.
2 Buy spot pounds with the borrowed dollars – thus tending to increase $S(\$/£)$.
3 Buy a pound security – thus tending to reduce $r_£$.
4 Sell the pound investment proceeds forward for US dollars – thus tending to reduce $F_{1/4}(\$/£)$.

Table 8.2 *Points off the interest-parity line*[a]

	Point					
	A	B	C	D	E	F
Interest differential	+.02	−.01	−.04	−.02	+.01	+.04
Forward premium[b]	−.04	−.01	+.02	+.04	+.01	−.02
Covered margin	−.02	−.02	−.02	+.02	+.02	+.02

Notes
a dollar advantage = +; dollar disadvantage = −.
b Forward premium (+) or discount (!) on the US dollar.

The inequality in equation (8.12) can be represented in Figure 8.4 by points such as A, B, and C that are above and to the left of the 45-degree line. The character of these points is summarized in Table 8.2. At point B, for example, dollar interest rates are 1 percent lower than pound interest rates, and at the same time the dollar is at a 1 percent forward discount. For both reasons there is an advantage to covered borrowing of dollars and investing in pounds. The **covered margin** or advantage of doing this is the interest differential ($r_£ - r_S$), plus the forward pound premium – on principal plus interest – for a total of 2 percent. In terms of equation (8.12), the inequality holds because the left-hand side is negative (−0.01) and the right-hand side is positive (+0.01). As we shall explain later, the covered interest arbitrage involving each of the four steps we have distinguished, will tend to restore covered interest parity by pushing the situation at B back towards the parity line. The same thing will occur at every other point off the interest-parity line. For example, consider the situation at point A, where there is an incentive to borrow dollars and invest in pounds, covered. The extra borrowing in dollars to profit from the arbitrage opportunity will put upward pressure on dollar interest rates. If the borrowing is through selling dollar money-market instruments, efforts to sell them will reduce their prices. For given coupons or maturity values, this will raise their yields. Thus, we will find r_S increasing. The increase in r_S can be represented in Figure 8.4 as a force pushing to the right of A, towards the interest-parity line.

The second step in covered interest arbitrage requires the spot sale of US dollars for British pounds. This will help bid up the spot price of the pound; that is, $S(\$/£)$ will increase. For any given $F_{1/4}(\$/£)$, the higher value of $S(\$/£)$ will reduce the numerator and increase the denominator and thereby reduce the value of the forward pound premium

$$\frac{F_{1/4}(\$/£) - S(\$/£)}{S(\$/£)}$$

This is shown in Figure 8.4 by an arrow pointing downward from A, towards the interest-parity line.

The pounds that were purchased will be used to invest in pound securities. Extra pound-security buyers will, *ceteris paribus*, cause the price of pound securities to increase, and therefore cause pound yields to decrease, that is, $r_£$ will fall. This means an increase in $(r_S - r_£)$, which is shown by an arrow pointing to the right from A. Again, the movement is back towards the covered interest-parity line.

Covering the pound proceeds of the investment by the forward sale of pounds will lower $F_{1/4}(\$/£)$. For every given value of $S(\$/£)$, there will be a lower value of the pound premium, $[F_{1/4}(\$/£) - S(\$/£)]/S(\$/£)$. Thus, there is a second force that will also push downward from A towards the parity line.[6] We can observe, of course, that since all four steps of arbitrage occur simultaneously, all the forces shown by the arrows occur simultaneously.

Points B and C in Figure 8.4, like point A, indicate profitable opportunities for covered borrowing in dollars and investment in pounds, and so there will be changes in interest and exchange rates also at these points as shown by the arrows. For example, at

6 There is an additional force pointing downward from A due to the lower pound interest rate. However, this is the effect of the reduced premium on the pound interest and is very small.

point C the dollar-interest rate is 4 percent lower than the pound rate. This is only partially offset by a 2-percent annual forward discount on pounds. This will encourage covered arbitrage flows towards pound investments. As before, there will be borrowing in dollars, and hence an increase in r_S; spot purchases of pounds, which will raise $S(\$/£)$; investment in pound securities which will lower $r_£$; and forward sales of pounds, which will lower $F_{1/4}(\$/£)$. All these changes are forces back to the interest-parity line. Indeed, at any point above the interest-parity line the forces shown by the arrows emanating from A, B, and C in Figure 8.4 are at work. We find that if we are off the covered interest-parity line and above it, market forces force us back down towards the line.

Below the interest-parity line, forces push us back up. At points such as D, E, and F, covered interest arbitragers will wish to borrow in pounds and invest in dollars. For example, at point E, dollar interest rates are 1 percent higher than pound rates, *and* the dollar is at a 1-percent forward premium. Thus dollar investments have a 2-percent advantage. This will cause arbitragers to sell pound securities, lowering their prices and raising $r_£$. This is shown in Figure 8.4 by an arrow that points to the left. The interest arbitragers then sell pounds for dollars, lowering $S(\$/£)$ and causing a movement upward, towards the parity line. They also purchase dollar securities, lowering r_S and causing a second movement towards the left. Hedging by buying pounds forward for dollars increases $F_{1/4}(\$/£)$, thereby raising the forward premium on sterling, $4[F_{1/4}(\$/£) - S(\$/£)]/S(\$/£)$. This means that again there will be movement towards the line, since the forward premium is the primary component on the vertical axis. (The other component, $r_£$, also causes movement, albeit small, back up towards the parity line.)

We find that above the covered interest-parity line, dollar borrowing and pound investment, with the associated spot and forward transactions, push us back towards the line. Below the line, pound borrowing and dollar investment with associated currency transactions also push us back towards the line. The amount of adjustment in interest rates versus spot and/or forward exchange rates depends on the "thinness" of the markets. For example, if a large part of the adjustment towards covered interest parity occurs in the forward rate, the paths followed from points such as A or E back towards the parity line will lie closer to the vertical arrows than to the horizontal ones. In such a case we can think of the forward premium/discount as being determined by the interest differential, rather than vice versa.[7]

COMBINING PPP AND INTEREST PARITY

The uncovered interest-parity condition

Equation (8.10) is the condition for *hedged* or *covered* interest-parity because it involves the use of the forward market. It can be argued that a similar *unhedged* interest-parity condition should also hold. This follows because, as we explained in Chapter 3 and shall confirm later, speculation will make the forward exchange rate approximately equal to the expected future spot rate. That is, if we define $S_n^*(\$/£)$ as in Chapter 3, namely

> $S_n^*(\$/£)$ is the expected spot exchange rate between the dollar and the pound in n years' time,

then it follows that to a close approximation

$$S_n^*(\$/£) = F_n(\$/£) \qquad (8.13)$$

Recall that the reason equation (8.13) holds is that if

$$S_n^*(\$/£) > F_n(\$/£)$$

speculators will buy pounds n-years forward; they can buy pounds forward for less than they expect to be able to sell them. This will force up the forward rate, $F_n(\$/£)$, until it is no longer less than the expected future spot rate. Similarly, if

$$S_n^*(\$/£) < F_n(\$/£)$$

7 Indeed, foreign exchange traders and brokers frequently compute forward premiums and discounts from interest differentials.

speculators will sell pounds n-years forward; they can sell forward pounds for more than they expect to be able to buy them when they honor their forward contract. Selling pounds forward pushes the forward rate down until it is no longer more than the expected future spot rate. Only when equation (8.13) holds is the forward rate in equilibrium in the sense that speculative pressures are not forcing the forward rate up or down.

Substituting equation (8.13) into equation (8.10) allows us to say that, to a close approximation, **uncovered interest parity** should hold in the form

$$(1 + r_\$)^n = \frac{S_n^*(\$/£)}{S(\$/£)}(1 + r_£)^n \qquad (8.14)$$

This is only an approximate condition because *uncovered* interest parity involves risk: we assume $S_n^*(\$/£) = F_n(\$/£)$ and as we shall see in Chapter 13, this assumption is invalid if there is a risk premium in the forward market.

Equation (8.14) can be put in a different form by noting that by definition

$$S_n^*(\$/£) \equiv S(\$/£)(1 + \dot{S}^*)^n \qquad (8.15)$$

where $\dot{S}^*$ is the average annual expected rate of change of the exchange rate. Substituting equation (8.15) into equation (8.14) gives

$$(1 + r_\$)^n = (1 + \dot{S}^*)^n(1 + r_£)^n \qquad (8.16)$$

Taking the nth root of both sides gives

$$1 + r_\$ = 1 + \dot{S}^* + r_£ + \dot{S}^* \cdot r_£ \qquad (8.17)$$

Assuming $\dot{S}^*$ and $r_£$ are small compared to 1, the "interaction term" $\dot{S}^* \cdot r_£$ will be very small, allowing us to write to a close approximation

$$r_\$ - r_£ = \dot{S}^* \qquad (8.18)$$

That is, the interest differential should approximately equal the expected rate of change of the spot exchange rate.

The expectations form of PPP

We recall from Chapter 7 that the expected dollar return from holding commodities in the United States is $\dot{P}_{US}^*$, that is, the expected US rate of inflation. Similarly, we recall that the expected dollar return from holding commodities in Britain is $\dot{P}_{UK}^* + \dot{S}^*$ because there are expected changes both in the pound prices of commodities and in the dollar value of the pound. We have seen that if we ignore risk, the rates of return from holding commodities in the two countries will be driven to equality by speculators until[8]

$$\dot{P}_{US}^* - \dot{P}_{UK}^* = \dot{S}^* \qquad (7.17)$$

where $\dot{S}^*$ is the expected rate of change of the spot rate, $S(\$/£)$. Equation (7.17) is the PPP condition in terms of expectations. It is known variously as the **expectations, speculative,** and **efficient markets PPP.**

The interrelationship of the parity conditions

If we take the PPP condition in its expectations form equation (7.17), and compare it with the uncovered interest-parity condition in equation (8.18), we note a clear similarity. We have

$$\dot{P}_{US}^* - \dot{P}_{UK}^* = \dot{S}^* \qquad (7.17)$$

$$r_\$ - r_£ = \dot{S}^* \qquad (8.18)$$

The right-hand sides of these two equations are equal. It follows that the left-hand sides must likewise be equal. This means that

$$r_\$ - r_£ = \dot{P}_{US}^* - \dot{P}_{UK}^* \qquad (8.19)$$

By rearranging, we have

$$r_\$ - \dot{P}_{US}^* = r_£ - \dot{P}_{UK}^* \qquad (8.20)$$

The two sides of this equation are the two currencies' interest rates less the expected rates of inflation in the associated two countries. The interest rate minus expected inflation is the expected **real interest rate,** popularized principally by Irving Fisher.[9] As a result, equation (8.20) is called the **Fisher-open condition.**[10] The Fisher-open condition states that the expected real rates of interest are equal in different countries. From purchasing-power-parity and uncovered interest parity we have been able to derive an equality relationship between expected real returns in different countries.[11]

The equality of expected real interest rates can be considered as having an independent existence, one that does not have to be derived from PPP and interest parity. It follows from investors allocating their funds to where expected real returns are highest. Investing according to the highest expected real yield will tend to reduce returns in the countries with high returns where funds are sent – because of the greater supply of funds. It will also tend to increase expected returns in countries from which the funds are taken – because of the reduced supply of funds. The flow of funds will continue until the expected real returns in different countries are equalized.[12]

8 As before, we assume no holding costs, or that holding costs are equal in the two countries.

9 See Irving Fisher, *The Theory of Interest*, A. M. Kelley, New York, 1965.

10 Generally, economists refer to equation (8.18), not equation (8.20), as the Fisher-open condition. However, since Fisher spoke of real interest rates as actual rates minus the expected inflation rate, equation (8.20) would be more appropriately referred to by his name. Equation (8.18) should perhaps be called the "interest-open" condition, where "open" means "open economy."

11 We state this in terms of countries rather than currencies because inflation refers to countries. Of course, returns within countries are in the countries' currencies.

12 The relationship between security yields in different countries can be extremely complex because of taxes, regulations, currency risks, citizens' tastes, and so on. The problem has been tackled by F. L. A. Grauer, R. H. Litzenberger, and R. E. Stehle, "Sharing Rules and Equilibrium in an International Capital Market under Uncertainty," *Journal of Financial Economics*, June 1976, pp. 233–56, and Fischer Black, "International Capital Market Equilibrium with Investment Barriers," *Journal of Financial Economics*, December 1974, pp. 337–52. Few data exist on real rates in different countries. See, however, Robert Z. Aliber, "Real Interest Rates in a Multicurrency World," unpublished paper, University of Chicago.

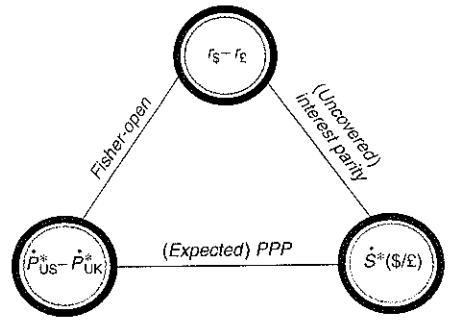


Figure 8.5 *The interdependence of exchange rates, interest rates, and inflation rates*

Notes
Interest parity, PPP, and the Fisher-open condition are related. Any one of these conditions can be derived from the other two.

If we write the uncovered interest parity, expected purchasing-power-parity, and Fisher-open conditions all together, that is,

interest parity: $r_\$ - r_£ = \dot{S}^*(\$/£)$

PPP: $\dot{P}_{US}^* - \dot{P}_{UK}^* = \dot{S}^*(\$/£)$

Fisher-open: $r_\$ - \dot{P}_{US}^* = r_£ - \dot{P}_{UK}^*$

we find that we can derive any one from the other two. This is left as an end-of-chapter problem for the reader. The conditions are shown in Figure 8.5. Each side of the triangle in Figure 8.5 represents a condition. The figure helps clarify why satisfying any two conditions implies that the remaining condition is satisfied.

Because each of the three parity conditions along the sides of Figure 8.5 can be derived from the other two, any one condition must be correct if the other two are correct. For example, if we believe expected real returns are equal in different countries and that uncovered interest parity holds precisely, we are implicitly accepting that PPP in its expectations form also holds precisely.

WHY COVERED INTEREST DIFFERENCES PERSIST

In reality, covered interest parity holds very closely, but it does not hold precisely. This is

apparent from, for example, the covered interest differentials in Table 8.1. The failure to achieve exact covered interest parity could occur because in actual financial markets there are

1 transaction costs;
2 political risks;
3 potential tax advantages to foreign exchange gains versus interest earnings; and
4 liquidity differences between foreign securities and domestic securities.

In addition, uncovered interest parity may not hold because of risk aversion: recall that uncovered parity is based on spot rate expectations, not on forward rates which eliminate foreign exchange risk.

In this chapter we explain how the four factors listed above contribute to departures from covered interest-parity. Later, in Chapter 14, we show that these are the same factors that influence cash management.

Transaction costs and interest parity

The cost of transacting in foreign exchange is reflected in the bid-ask spread in exchange rates. The bid-ask spread represents the cost of two foreign exchange transactions, a purchase and a sale of foreign currency. That is, if a person buys and then immediately sells a foreign currency, the cost of these two transactions is the difference between the buying and selling prices of the currency, which is the bid-ask spread. Covered investment or borrowing involve two foreign exchange transaction costs – one on the spot market and the other on the forward market. These two transaction costs discourage foreign-currency denominated investment and borrowing. Interest arbitrage also involves two foreign exchange transaction costs, since the borrowed currency is sold spot and then bought forward. However, there are additional transaction costs of interest arbitrage due to interest-rate spreads. This is because the borrowing interest rate is likely to exceed the investment interest rate.[13]

It might seem that the extra cost of investing in foreign currency-denominated securities vis-à-vis domestic-currency securities would require covered foreign-currency yields to be higher than domestic-currency yields before investors choose the foreign-currency alternative; investors need to find it worthwhile to incur the extra transaction costs for a foreign-currency security. Similarly, it might seem that borrowing costs in foreign currency would have to be lower than borrowing costs in domestic currency before borrowers choose the foreign-currency alternative; borrowers face extra costs when borrowing via the foreign-currency swap. In other words, it might seem that there could be deviations from interest parity by up to the extra transaction costs of investing or borrowing in foreign currency before the benefits of the foreign alternative are sufficient to compensate for the added costs. In terms of Figure 8.4, it would appear that transaction costs could allow the situation to be slightly off the interest-parity line; the apparent advantages to foreign-currency investments/borrowing at points just off the line do not trigger foreign borrowing/investing because the benefits are insufficient to compensate for the costs. Indeed, because the cost of covered interest arbitrage includes the borrowing – investing interest rate spread as well as foreign exchange transaction costs, it might appear that deviations from interest parity could be relatively large before being sufficient to compensate for the transaction costs of covered interest arbitrage.

Despite the preceding, which would suggest that deviations from interest parity could result from transaction costs, it has generally become recognized that transaction costs do not contribute to deviations from interest parity. A major reason for this recognition is the realization that one-way

13 The borrowing-investment spread can be considered as a transaction cost in the same way that we consider the bid-ask spread on currencies a transaction cost, namely, it is the cost of borrowing and then immediately investing the borrowed funds.

interest arbitrage circumvents transaction costs in foreign exchange and securities markets. We can explain the nature of one-way interest arbitrage and how it influences the interest-parity condition by contrasting one-way and **round-trip arbitrage**. In order to do this, we need to be explicit about foreign exchange and borrowing – lending transaction costs. Using the same notation as in Chapter 2 for transaction costs on spot exchange, let us use the following definitions:

> $S(\$/\text{ask}£)$ and $F_n(\$/\text{ask}£)$ are respectively the spot and n-year forward exchange rates when buying pounds with dollars, and $S(\$/\text{bid}£)$ and $F_n(\$/\text{bid}£)$ are respectively the spot and forward exchange rates when selling pounds for dollars. $r^I_\$$ and $r^I_£$ are the interest rates earned on investments in the two currencies, and $r^B_\$$ and $r^B_£$ are the interest rates on borrowing in the two currencies.

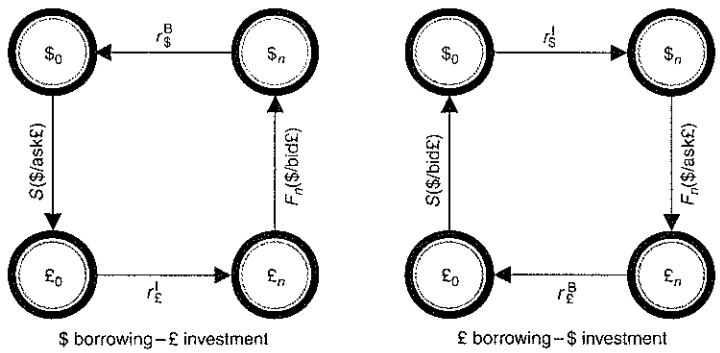
This notation is used in Figure 8.6 to show the difference between one-way and round-trip arbitrage and the implications of this distinction for covered interest parity.

Figure 8.6a illustrates round-trip covered interest arbitrage. As before, the four corners of the diagram show current dollars ($\$_0$), current pounds, ($£_0$), future dollars ($\$_n$), and future pounds ($£_n$). The arrows drawn between the corners of the figure show the interest rates or exchange rates when going between the corners in the directions of the arrows. For example, when going from $\$_0$ to $£_0$ as shown by the downward-pointing arrow in the left-hand panel of Figure 8.6a, the transaction occurs at the spot exchange rate, $S(\$/\text{ask}£)$: pounds are being purchased. Similarly, when going from $\$_n$ to $\$_0$ as shown by the upper, horizontal, leftward-pointing arrow in the left-hand panel of Figure 8.6a, this involves borrowing dollars and so occurs at the dollar-borrowing rate, $r^B_\$$.

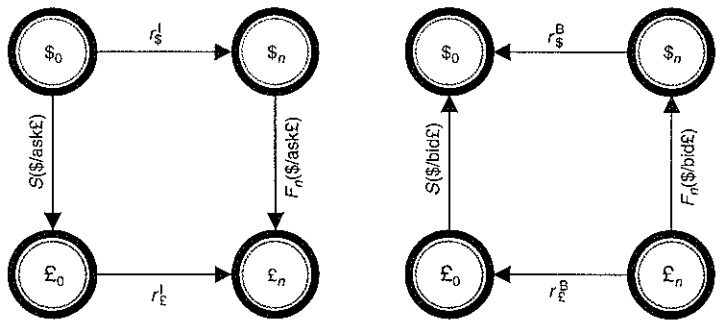
The left-hand diagram in Figure 8.6a shows round-trip arbitrage involving borrowing in dollars and investing in pounds. To understand the nature of this arbitrage we begin at corner $\$_n$. The top leftward-pointing arrow from $\$_n$ shows the interest rate on dollar borrowing which gives immediate dollars, $\$_0$, in return for paying dollars back in the future, $\$_n$. The left downward-pointing arrow shows the spot exchange rate at which the borrowed dollars are exchanged into pounds; the pounds must be purchased, so the spot rate is the ask rate for pounds, $S(\$/\text{ask}£)$. The bottom rightward-pointing arrow shows the interest rate earned on the pound-denominated investment which converts today's pounds, $£_0$, into future pounds $£_n$. Finally, the right upward-pointing arrow shows the forward exchange rate at which the dollars needed for repaying the dollar loan are purchased with pounds, $F_n(\$/\text{bid}£)$. The counterclockwise journey in this left-hand diagram in Figure 8.6a from $\$_n$ and back to $\$_n$ is seen to involve foreign exchange transaction costs – the ask on spot pounds $S(\$/\text{ask}£)$ versus bid on forward pounds $F_n(\$/\text{bid}£)$, and therefore a bid-ask spread -- plus borrowing – investment transaction costs – the borrowing rate on dollars $r^B_\$$ versus the investment rate on pounds, $r^I_£$. That is, round-trip arbitrage is expensive in terms of facing costs in the currency and in the security markets.

The right-hand diagram in Figure 8.6a illustrates the alternative direction of round-trip arbitrage, with borrowing of pounds and investment in dollars. Starting at $£_n$, pounds are borrowed at $r^B_£$ giving the borrower current pounds $£_0$. These are sold spot for dollars at $S(\$/\text{bid}£)$ and the dollars invested at $r^I_\$$. The pound-denominated loan is covered by buying forward pounds at the forward ask rate for pounds, $F_n(\$/\text{ask}£)$. As with the left-hand figure in Figure 8.6a, we see that the pound borrowing-dollar investment arbitrage also involves a transaction cost spread in the foreign exchange market – spot bid versus forward ask on pounds – and in the securities market – pound borrowing rate versus the dollar investment rate.

If the maximum possible sizes of deviations from covered interest parity due to transaction costs were determined only by round-trip covered interest arbitrage, the deviations could be quite large. This is because for round-trip interest arbitrage to be profitable it is necessary to overcome the



(a) *Round-trip covered interest arbitrage*



(b) *One-way covered interest arbitrage*

Figure 8.6 *One way and round-trip interest arbitrage*

Notes
Round-trip interest arbitrage, illustrated in (a), involves going along all four sides of the diagram at the interest rates and exchange rates that are shown. The presence of four transaction costs allows for large deviations from interest parity. One-way interest arbitrage, illustrated in (b), involves comparing two alternative ways of going from one corner to another corner that is diagonally opposite. Either route involves two transaction costs, but since these are for the same types of transactions, they do not cause deviations from interest parity.

transaction costs in the foreign exchange markets and the borrowing – lending spread on interest rates. Let us attach some numbers to see the size of deviations that might result. We shall use transactions costs faced by the lowest cost arbitragers, such as large commercial banks, since it is they who are likely to act first and preclude others from profiting from interest arbitrage.

Let us assume a potential interest arbitrager can borrow for 1 year at $\frac{1}{4}$ percent (25 basis points) above his or her investment rate, and can borrow a sufficient amount to reduce the spot and forward transaction costs both to only $\frac{1}{10}$ of 1 percent. In this situation, it is necessary that the interest-parity deviation calculated using interest rates and exchange rates that exclude transaction costs would have to be almost $\frac{1}{2}$ of 1 percent for profitable arbitrage. This is because it is necessary to earn $\frac{1}{4}$ of one percent to cover the borrowing – investment spread, and another $\frac{1}{5}$ of 1 percent to cover the two foreign exchange transaction costs, those for the spot and forward exchange transactions, both being

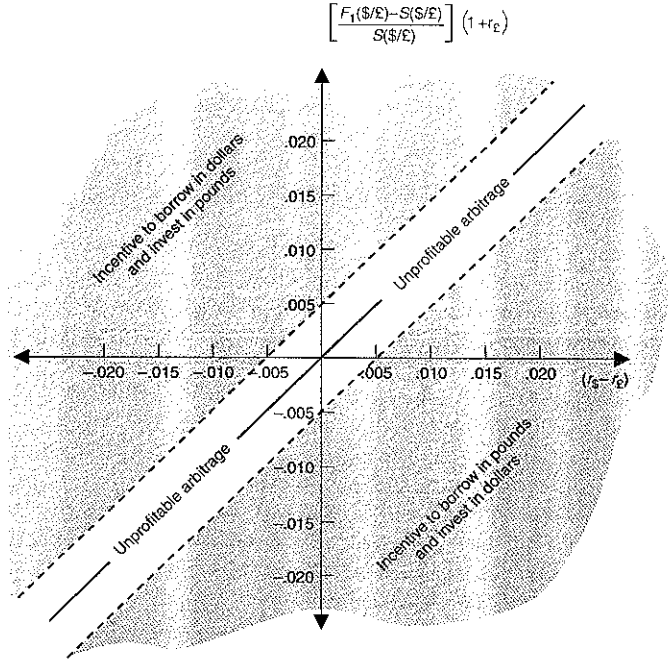


Figure 8.7 *Interest parity in the presence of transaction costs, political risk, or liquidity premiums*

Notes
Interest parity might not hold exactly because of transaction costs, political risk, and liquidity preference. This means interest rates and exchange rates may not plot on the interest-parity line. Rather, they may be somewhere within a band around the line; only outside this band are the covered yield differences enough to overcome the costs and/or political risk of covered interest arbitrage. However, the band is narrow because there are some participants for whom the costs and political risk of arbitrage are unimportant or irrelevant. For example, transaction costs are irrelevant for one-way arbitragers.

$\frac{1}{10}$ of 1 percent. This is illustrated in Figure 8.7. We can interpret the interest rates on the horizontal axis as the midpoints between the borrowing and lending rates, and the exchange rates on the vertical axis as the midpoints between the "bid" and "ask" rates.

We show a band around the interest-parity line within which round-trip interest arbitrage is unprofitable. This band has a width of approximately $\frac{1}{2}$ percent on either side of the interest-parity line, as is seen, for example, along the horizontal axis around the origin. The band reflects the fact that arbitrage must cover the $\frac{1}{2}$ percent lost in transaction costs/spreads. Above and to the left of the band it is profitable to borrow in dollars and make covered investments in pounds, and to the right of the band it is profitable to do the reverse.[14]

14 If interest rates were for 3 months rather than 1 year, the band would be considerably wider. Even if the borrowing – lending spread remained at $\frac{1}{4}$ of 1 percent and the foreign exchange transaction costs remained at $\frac{1}{10}$ of 1 percent, the band would be more than one full percent on either side of the interest-parity line. The reason is that the costs of buying spot and selling forward are incurred within a 3-month period, and when annualized are effectively 4 times larger. With $\frac{1}{5}$ percent lost in 3 months, it is necessary for the interest arbitrage to generate $\frac{4}{5}$ of a percent to cover foreign-exchange costs, plus $\frac{1}{4}$ of a percent to cover the borrowing–lending spread. If we were dealing with 1-month or shorter arbitrage, the potential deviations from interest parity would be even larger.

Let us next consider the implication of *one-way* arbitrage.

One-way arbitrage can come in various forms. However, we need to consider only the form which involves the lowest transaction cost, because it is this arbitrage that will determine the maximum deviation from interest parity; just as the *arbitrager* with the lowest cost of interest arbitrage drives interest rates and exchange rates to levels closest to the interest-parity line, so it is that the *form* of one-way arbitrage that faces the lowest transaction cost drives us closest to the parity line.

Let us consider first the one-way interest arbitrage illustrated by the left-hand diagram in part b of Figure 8.6. This shows an arbitrager holding dollars ($\$_0$) who wants pounds in the future ($£_n$), perhaps to pay for something purchased from Britain.[15] The arbitrager has two choices to go from current dollars to future pounds. The dollars can be sold for pounds on the spot market and invested in pound-denominated securities until the pounds are needed, or the dollars can be invested in dollar securities and forward pounds can be purchased. The first of these choices is illustrated by the downward-pointing arrow on the left of the diagram from $\$_0$ to $£_0$ and the rightward-pointing arrow along the bottom from $£_0$ to $£_n$. The second choice is illustrated by the upper rightward-pointing arrow from $\$_0$ to $\$_n$ and the downward-pointing arrow on the right from $\$_n$ to $£_n$. We note that both routes take the arbitrager from today's dollars, $\$_0$, to future pounds, $£_n$. The choice is to pick the route providing the future pounds at lower cost.

15 The use of the term "arbitrager" in this context stretches the usual meaning of the term, because we have assumed that dollars are already held, and that there is already a need for pounds in the future. Without engaging in excessive semantics we can note that the essence of what our "arbitrager" is doing is judging which of two ways of going from current dollars to future pounds is the cheaper. This choice of the preferred route is the same type of choice met in deriving cross exchange rates in Chapter 2. The term "arbitrage" comes from "arbitrate," which means "choose." However, a reader who still objects to the term "arbitrager" can substitute his or her own term; the conclusion is the same whatever word we use.

If the choice that is made is to invest in dollar securities and buy pounds forward, each pound purchased will cost $\$F_n(\$/\text{ask}£)$. In order to have this many dollars in n-years requires investing today at $r^I_\$$

$$\$\frac{F_n(\$/\text{ask}£)}{(1+r^I_\$)^n} \tag{8.21}$$

This is the cost per pound using the dollar investment, forward pound route from $\$_0$ to $\$_n$. If the other choice is made, that is, to buy pounds immediately and invest in pound-denominated securities for n years, the number of pounds that must be purchased immediately for each pound to be obtained in n years is

$$£\frac{1}{(1+r^I_£)^n} \tag{8.22}$$

The cost of this number of pounds at the spot price of pounds, $S(\$/\text{ask}£)$, is:

$$\$\frac{S(\$/\text{ask}£)}{(1+r^I_£)^n} \tag{8.23}$$

In the same way that we saw that there are forces pushing interest rates and exchange rates to the interest-parity line, there are forces making the amounts in equations (8.21) and (8.23) equal to each other. For example, if equation (8.23) gave a lower cost per pound than equation (8.21), there would be spot purchases of pounds and funds invested in pound-denominated securities. These actions would increase $S(\$/\text{ask}£)$ and reduce $r^I_£$. At the same time, the lack of forward pound purchases would reduce $F_n(\$/\text{ask}£)$, and the capital outflow from the United States would increase $r^I_\$$. The choice between alternatives would therefore drive them to the same cost, that is,

$$\frac{F_n(\$/\text{ask}£)}{(1+r^I_\$)^n} = \frac{S(\$/\text{ask}£)}{(1+r^I_£)^n} \tag{8.24}$$

We can rewrite (8.24) as

$$(1+r^I_\$)^n = \frac{F_n(\$/\text{ask}£)}{S(\$/\text{ask}£)}(1+r^I_£)^n \tag{8.25}$$

If the sizes of transaction costs in the spot and forward foreign exchange markets were the same, then equation (8.25) would plot as a 45-degree line through the origin in the interest-parity diagram, Figure 8.7. This is because we have investment interest rates on both sides of the equation, and we have "ask" exchange rates for both the forward and spot transactions.[16] This means that if spot and forward transaction costs were equal, the choice we have described would drive us all the way to the interest-parity line even though there are transaction costs. The reason this happens is that either method of going from $\$_0$ to $£_n$ requires buying pounds, the only difference being whether they are purchased on the spot or the forward market. Therefore, transaction costs will be paid whatever choice is made. Similarly, investment interest rates are earned if the choice is to buy pounds spot or forward, the only difference being in which currency the interest is earned.

Another form of one-way interest arbitrage is illustrated by the lower right-hand diagram in Figure 8.6b. The choice in this case is between two ways of going from $£_n$ to $\$_0$. (A US exporter who is to receive pounds and needs to borrow dollars would be interested in going from $£_n$ to $\$_0$.) The two ways of going from the $£_n$ to $\$_0$ involve either going from $£_n$ to $£_0$ by borrowing pounds, and then from $£_0$ to $\$_0$ via the spot market, or going from $£_n$ to $\$_n$ via the forward market, and from $\$_n$ to $\$_0$ by borrowing dollars.

If the route that is taken from $£_n$ to $\$_0$ is to borrow pounds and sell them spot for dollars, the number of pounds that can be borrowed today for each pound to be repaid in n years is

$$£\frac{1}{(1+r^B_£)^n}$$

This number of pounds, when sold spot for dollars, provides

$$\$\frac{S(\$/\text{bid}£)}{(1+r^B_£)^n} \tag{8.26}$$

The alternative route of selling pounds forward and borrowing dollars means receiving in the future for each pound sold

$$\$F_n(\$/\text{bid}£)$$

The number of dollars that can be borrowed today and repaid with these pounds is

$$\$\frac{F_n(\$/\text{bid}£)}{(1+r^B_\$)^n} \tag{8.27}$$

The dollar amounts in equations (8.26) and (8.27) show the dollars available today, $\$_0$, for each pound in the future, $£_n$. The choice between the alternative ways of obtaining dollars will drive exchange rates and interest rates to the point that

$$\frac{S(\$/\text{bid}£)}{(1+r^B_£)^n} = \frac{F_n(\$/\text{bid}£)}{(1+r^B_\$)^n}$$

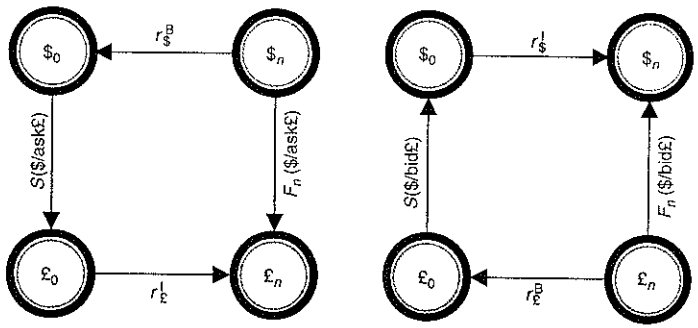
that is,

$$(1+r^B_\$)^n = \frac{F_n(\$/\text{bid}£)}{S(\$/\text{bid}£)}(1+r^B_£)^n \tag{8.28}$$

Again, if forward and spot transaction costs are equal, this is an exact interest-parity line; we have borrowing interest rates on both sides, and both exchange rates are bid rates.

In fact, forward exchange transaction costs are higher than spot costs – recall from Chapter 3 that forward spreads are wider than spot spreads – and so equation (8.25) and equation (8.28) might differ a little from the interest-parity line drawn without transaction costs. However, the departures

16 That is, because both exchange rates are "ask" rates, the exchange rates in the numerator and denominator of equation (8.25) are both on the high side of the bid-ask spread. To the extent that transaction costs in the spot and forward markets are equal, they cancel. Similarly, because the interest rates on the two sides of equation (8.25) are investment rates, they are both on the low side of the borrowing-lending spread. The cost-component again cancels.



(a) *Pounds needed in future* (b) *Need to sell pounds*

Figure 8.8 *A more roundabout one-way arbitrage*

Notes
We can reduce the band around the interest-parity line by considering whether to use the forward market or the spot market plus money markets. This choice, however, involves one transaction versus three transactions and is likely to leave larger deviations from interest parity than other, more similar one-way arbitrage choices.

will be much smaller than those obtained from consideration of round-trip interest arbitrage.[17] This is because round-trip interest arbitrage involves the borrowing – investment interest-rate spread and foreign exchange transaction costs of buying spot and selling forward, or of buying forward and selling spot. On the other hand, one-way arbitrage does not involve interest-rate spreads, and foreign exchange transaction costs are faced whatever choice is made.

The one-way arbitrage we have described produces the interest-parity line because we have established situations where the arbitrager has in any case to buy/sell foreign exchange and to invest/borrow. An alternative one-way arbitrage is a choice between buying/selling forward on the one hand, and buying/selling spot and using the money markets on the other hand.[18] For example an arbitrager could buy pounds forward, or alternatively could borrow dollars, buy pounds spot, and invest the pounds. Either way the arbitrager receives pounds in the future and has to deliver dollars. This is illustrated in Figure 8.8a, where the arbitrager can go from $\$_n$ to $£_n$ via the forward market (the downward-pointing arrow on the right-hand side), or by borrowing dollars (the upper leftward-pointing arrow), buying pounds spot with the borrowed dollars (the left-hand downward-pointing arrow), and investing in pounds (the lower rightward-pointing arrow). This one-way arbitrage does make *foreign exchange* transaction costs irrelevant, or at least less relevant, because it involves ask rates in either case. However, it does not avoid the *borrowing – investing* spread. It does not, therefore, take us as close to the interest-parity line as the type of one-way arbitrage we considered. We reach the same conclusion if we consider the reverse forward exchange, going from $£_n$ to $\$_n$, versus the

17 In fact, even if there are transaction cost differences between spot and forward exchange, when all one-way arbitrages are considered simultaneously with a requirement that there is both supply and demand in every market, interest parity holds exactly. This is shown in Maurice D. Levi, "Non-Reversed Investment and Borrowing, Transaction Costs and Covered Interest Parity," *International Review of Economics and Finance*, 2, 1992, pp. 107–19.

18 This is the more usual one-way arbitrage considered in explaining why transaction costs are not important. For example, see Alan V. Deardorff, "One-Way Arbitrage and its Implications for the Foreign Exchange Markets," *Journal of Political Economy*, April 1975, pp. 351–64.

alternative of borrowing pounds, using these to buy dollars spot, and investing the dollars. This is illustrated in Figure 8.8b; it again gives a line further from the interest-parity line than the one-way arbitrage we described.

An alternative way of concluding that the deviations from interest parity as a result of transaction costs are small is to consider the choice faced by third-country borrowers and investors. For example, if Japanese or German investors and borrowers are looking for the best currency to invest or borrow, they will drive the situation between dollars and pounds to the interest-parity line. This is because the Japanese or Germans pay foreign exchange costs whatever the currency of their investment or borrowing, and compare investment rates in the two currencies, or borrowing rates in the two currencies.[19] For this reason, or because of the presence of one-way arbitrage, we can expect deviations from interest parity to be too small for round-trip arbitrage to ever occur. We conclude that transaction costs are probably not a cause of deviations from interest parity.[20]

Political risk and interest parity

When securities denominated in different *currencies* trade in different *countries*, deviations from interest parity can result from **political risk**. Political risk involves the uncertainty that while funds are invested in a foreign country, they may be frozen, become inconvertible into other currencies, or be confiscated. Even if such extremes do not occur, investors might find themselves facing new or increased taxes in the foreign country. Usually, the investment that involves the least political risk is at home; if funds are invested abroad, to the risk of tax or other changes at home is added the risk of changes in another political jurisdiction. However, it is possible that for investors in some countries, it is politically less risky to send funds abroad. This will be true if investors thereby avoid politically risky possibilities at home. For example, people in some volatile countries have invested in Switzerland and the United States for political safety. In these circumstances, a foreign investment might be made even at a covered interest disadvantage. In general, however, we expect investors to require a premium from a foreign investment versus a domestic investment.

In diagrammatic terms, political risk creates a band like that shown in Figure 8.7; only in the area beyond some covered differential is there an incentive to invest abroad.[21] The political risk band does not have to be of equal width on the two sides of the interest-parity line if one country is viewed as riskier than the other. For example, Canadian yields are generally a little higher than US yields, even after allowance for forward hedging. This can be attributed to US investors viewing Canada as being politically more risky than Canadians view the US, thereby causing a larger political-risk premium on Canadian securities than on US securities.

It is important to remember that political risk relates to the country, not the currency, of investment. For example, there is no political risk involved in the deviations from covered interest-parity in Table 8.1 because the yields are assumed to be on different currency bank deposits in London. That is, there are no political differences between the instruments in Table 8.1, but rather only differences in the currencies of denomination. Indeed, by comparing yield differences when there are no political differences – as on bank deposits (covered) in a given country – with yield differences when there are

19 This assumes that cross exchange rates of pounds for yen or pounds for euros do not have a larger bid-ask spread than direct yen-dollar or euro-dollar exchange rates. When cross-exchange-rate spreads are larger, dollar investments and borrowing will be favored.

20 The arguments we have given for transaction costs to be a minor or irrelevant reason for deviation from interest parity can be supplemented by an argument advanced by Kevin Clinton involving the trading of swaps. See Kevin Clinton, "Transaction Costs and Covered Interest Arbitrage: Theory and Evidence," *Journal of Political Economy*, April 1988, pp. 358–70.

21 Of course, political risk does not create a band via the *borrowing* decision. Borrowers have to pay back their loans whatever happens in the country from which they borrowed.

political differences – as on Treasury bills (covered) in different countries – it is possible to estimate the importance of political risk. The extra spread when political risk is faced versus when it is not faced is a measure of the political risk premium.[22]

Even when covered yields are on instruments which trade in different countries, third-country investors might force interest rates and exchange rates onto the interest-parity line. For example, if Japanese investors view the United States and Britain as equally risky politically from their perspective, then they will drive the interest rates and exchange rates for the United States and Britain onto the interest-parity line. This is true even if investors in the United States and Britain perceive foreign investment as riskier than investment at home. Of course, if conditions are driven onto the interest-parity line, this will encourage US and British investors to keep funds at home, because neither is receiving compensation for the perceived risk of investing in the other country.

Taxes and interest parity

If taxes are the same on domestic and on foreign investment and borrowing, then the existence of taxes will make no difference in our investment and borrowing criteria or the interest-parity line; taxes will cancel out when yield comparisons are made. However, if tax rates depend on the country in which funds are invested or borrowed, the interest-parity condition will be affected. There are two ways in which taxes could conceivably affect the parity condition. One way involves withholding taxes, and the other involves differences between the tax rate on income and that on capital gains. Let us consider these in turn.

Withholding taxes

One might think that a potential cause of higher taxes on foreign earnings than on domestic earnings, and hence a band around the interest-parity line, is the **foreign resident withholding tax**. A withholding tax is a tax applied to foreigners at the source of their earnings. For example, when a Canadian resident earns $100 in the United States, the payer of that $100 is required to withhold and remit 15 percent of the earnings to the US Internal Revenue Service. Similarly, the earnings of US residents in Canada are subject to a withholding tax. Withholding taxes, however, are unlikely to offer a reason for a band around the covered interest-parity line.

As long as the rate of withholding is less than or equal to the tax rate that would be applied to the earnings at home, domestic **withholding tax credits** that are designed to avoid double taxation of income will offset the tax withheld. For example, suppose that a resident of the United States pays the equivalent of $15 on $100 of interest or dividends earned in Canada, and the total tax payable on the $100 when declared in the United States is $25. The Internal Revenue Service will grant the US resident a $15 credit on taxes paid to the taxing authority in Canada. Only an additional $10 will be payable in the United States. The investor ends up paying a total of $25, which is the same as she would have paid on $100 earned at home. Complete or full withholding tax credit leaves no incentive to choose domestic securities rather than foreign securities. Only if withholding tax credits are less than the amounts withheld will there be a reason to keep money at home.[23] This means that the interest-parity condition is in general not affected, and we have no band around the parity line as a result of withholding taxes.

Capital gains versus income taxes

Taxes can affect the investment and borrowing criteria and the interest-parity condition if investors pay different effective tax rates on foreign exchange earnings than on interest earnings. This can be the situation for investors who infrequently buy or sell foreign exchange. This is because such investors can obtain capital-account treatment of their foreign exchange gains or losses; gains and losses are normally given capital-account treatment if they are not part of the "normal conduct of business."[24] If the tax rate on capital gains is lower than that on ordinary income, this affects the slope of the interest-parity line. Let us see how, by considering a US investor who pays a lower effective tax rate on capital gains than on interest income.[25]

Let us write the US investor's tax rate on capital gains as τ_K and the US tax rate on income as τ_Y, and let us assume that for this particular investor, $\tau_Y > \tau_K$. Since all interest earnings are considered to be income, after paying taxes and ignoring transaction costs the US investor will receive from each dollar invested in dollar-denominated securities for 1 year

$$1 + (1 - \tau_Y)r_\$ \tag{8.29}$$

That is, the investor will lose a fraction τ_Y of the interest earned. If he or she instead invests in pound securities, then before taxes the US dollar receipts will be

$$\frac{F_1(\$/£)}{S(\$/£)}(1 + r_£) = \left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right] \times (1 + r_£) + (1 + r_£)$$

or

$$\frac{F_1(\$/£)}{S(\$/£)}(1 + r_£) = 1 + r_£ + \left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right](1 + r_£)$$

We have expanded the total dollar payback into three parts; the principal; the interest earned on the pounds; and the earnings/losses from exchange rate premiums/discounts on the principal and interest, $(1 + r_£)$. After taxes, if capital gains taxes are paid on foreign exchange earnings, even when hedged, the investor will receive only $(1 - \tau_Y)$ of the interest and $(1 - \tau_K)$ of the gain from the forward premium, that is,

$$1 + (1 - \tau_Y)r_£ + (1 - \tau_K) \times \left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right](1 + r_£) \tag{8.30}$$

We have used the income tax rate τ_Y on $r_£$, since all interest, whatever the currency or country of source, is subject to that rate.

We can show the effect of taxes in terms of the graphical presentation of interest parity if we proceed the same way as we did when we included transaction costs. The US investor for whom we have assumed $\tau_Y > \tau_K$ will be indifferent between investing in dollar or pound securities if the amounts in equations (8.29) and (8.30) are equal. This requires that

$$r_\$ - r_£ = \frac{1 - \tau_£}{1 - \tau_Y}\left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right] \times (1 + r_£) \tag{8.31}$$

In comparing equation (8.31) with the equation for the interest-parity line in Figure 8.4, which is

$$r_\$ - r_£ = \left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right](1 + r_£)$$

we see that differential taxes on income versus capital gains/losses adds $(1 - \tau_K)/(1 - \tau_Y)$ to the front of the forward premium/discount term. When the capital gains tax rate is lower than the income tax rate,

$$\frac{1 - \tau_K}{1 - \tau_Y} > 1$$

22 See Robert Z. Aliber, "The Interest Parity Theorem: A Reinterpretation," *Journal of Political Economy*, November/December 1973, pp. 1451–9.

23 Even when full credit is obtained, interest earnings are lost on the funds withheld in comparison with what might have been earned if taxes had been paid at home at the end of the tax period. This should, however, be a relatively small consideration except when interest rates are very high.

24 Even where the tax rate on realized capital gains is the same as on ordinary income, the *effective* tax rate on capital gains is lower if capital gains can be deferred, or if there are capital losses against which the capital gains can be taken. In many countries the *actual* tax rate on capital gains is lower than that on ordinary income.

25 For the conditions for capital-account treatment of foreign exchange earnings, see Martin Kupferman and Maurice D. Levi, "Taxation and the International Money Market Investment Decision," *Financial Analysts Journal*, July/August 1978, pp. 61–4.

For example, if $\tau_K = 0.10$ and $\tau_Y = 0.25$, then

$$\frac{1-\tau_K}{1-\tau_Y} = 1.20$$

This means that the line of indifference for investors who face lower taxes on foreign exchange earnings than on interest income is flatter than the 45-degree interest-parity line in Figure 8.4; each percent change in $[F_1(\$/£) - S(\$/£)]/[S(\$/£)] \times (1 + r_£)$ on the vertical axis in Figure 8.4 is associated with more than a 1 percent change in $r_\$ - r_£$ on the horizontal axis.

While some investors may enjoy a lower tax rate on foreign exchange than on interest earnings, banks and other major financial market players do not; such investors, for whom international investment is part of their normal business, pay the same tax on interest and foreign exchange earnings. For this reason we can expect interest rates and exchange rates to remain on the 45-degree interest-parity line; banks will take positions until the situation is on the line. This implies that those investors who *do* pay lower taxes on foreign exchange earnings than on interest income may find valuable **tax arbitrage** opportunities. For example, suppose interest rates and exchange rates are such that interest parity holds precisely on a before-tax basis with:

$$r_\$ = 8\%$$

$$r_£ = 4\%$$

$$\left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right](1 + r_£) = 4\%$$

For a US investor for whom $\tau_Y = 0.25$ and $\tau_K = 0.10$, US investments yield $(1 - \tau_Y)r_\$ = 6$ percent after tax, while covered pound investments yield after tax:

$$(1-\tau_Y)r_£ + (1-\tau_Y)\left[\frac{F_1(\$/£) - S(\$/£)}{S(\$/£)}\right](1 + r_£) = 6.6\%$$

The pound investment will be preferred on an after-tax basis even though interest parity holds on a before-tax basis. More generally, if covered interest-parity holds on a before-tax basis, investors with favorable capital gains treatment will prefer investments denominated in currencies trading at a forward premium. It is a natural extension of our argument to show that in the same tax situation, borrowers will prefer to denominate borrowing in currencies at a forward discount.[26]

Liquidity differences and interest parity

The liquidity of an asset can be judged by how quickly and cheaply it can be converted into cash. For example, when a *domestic*-currency denominated asset such as a 90-day security is sold before maturity after only 50 days, domestic security selling transaction costs must be paid that would not have been incurred had the security been held to maturity. If, however, a covered *foreign*-currency 90-day investment is sold after only 50 days, more than security selling costs are faced. This should be explained carefully.

The brokerage costs for selling a foreign-currency denominated security are likely to be similar to those for selling a domestic-currency security. However, transaction costs are faced when investors convert on the spot exchange market, the foreign exchange received from the sale of the foreign-currency security. These costs would have not been faced had the security been held to maturity and the proceeds converted according to the original forward contract. Further, when a foreign-currency denominated security is sold prior to maturity and the funds are converted into domestic currency, there is still the matter of honoring the original forward contract to sell the foreign exchange at the maturity of the foreign-currency investment. If cash managers want to avoid the foreign exchange risk that would be faced by leaving the original forward contract in effect, they must cover their position. In our example, if there is a 90-day forward contract and the funds are converted into domestic currency after only 50 days, the investor should buy a new forward exchange contract of 40 days maturity. This purchase of forward foreign exchange will offset the sale of foreign exchange that was part of the original covered investment. At the conclusion of the full 90-day period, the foreign exchange that was originally sold forward will be obtained from that bought forward 40 days previously.

The extra spot and forward exchange transaction costs from the premature sale of a foreign-currency investment require an initial advantage of covered foreign investment to make the initial investment worthwhile. There is also uncertainty at the time of the original investment concerning what the spot rate will be if redemption is early and what the rate will be for offsetting the original forward contract with a reverse contract. The transaction costs and uncertainty make foreign covered investments less liquid than domestic investments. **Liquidity preference** is hence another reason for a band around the covered interest-parity line. The amount of extra required return and hence the potential width of the band due to liquidity preference depend on the likelihood that the funds will be needed early, and on whether these funds can be borrowed by using the original covered investment to secure the loan. Since the required extra return depends on the *likelihood* that the funds will be needed, it is clear that this liquidity consideration is different from the transaction costs consideration discussed earlier, which involved *known* amounts of transaction costs. Liquidity does relate to transaction costs, but these are *expected* costs. Clearly, if it is known that funds will not be required, or if it is known that the foreign investment can be used as the guarantee or security for borrowing funds, no foreign yield premium is required. The more uncertainty there is concerning future needs and alternative sources of short-term financing, the higher are the premiums that should be required before venturing into foreign-currency securities. This will mean wider bands around the interest-parity line.[27]

As in the case of transaction costs and political risk, the choice by third-country investors of which currency-denominated securities to invest in should be symmetrical as far as liquidity preference is concerned. That is, if there is the same probability of needing to liquidate investments from either currency and the same transaction costs if liquidation occurs, liquidity preference in the presence of third-country investors should not cause deviations from interest parity.

Effect of the reasons for interest disparity on investment and borrowing

Each investor or borrower must evaluate yields and borrowing costs from his or her own perspective. This means using exchange rates that include transaction costs and consideration of applicable tax rates, and then comparing yield or borrowing-cost differences with the difference the investor or borrower believes is necessary to compensate for the risk or illiquidity that is faced. What we have argued suggests that, transaction costs are likely to work towards keeping investing and borrowing in the home currency. Withholding taxes are likely to matter only if withholding rates are higher than domestic tax rates, but differential taxes on interest income versus capital gains could induce investors with favorable capital gains treatment to place funds in currencies at a forward premium. As for political risk, this will create yield and cost differences that are not exploitable for those facing the political risk, although the differences may be exploitable by

26 This is because the high interest rates they pay are tax-deductible. For more on taxes and interest parity see Maurice D. Levi, "Taxation and 'Abnormal' International Capital Flows," *Journal of Political Economy*, June 1977, pp. 635–46.

27 It might be felt that movements in security values because of changes in market interest rates also affect the liquidity of domestic investment versus foreign investment. Although the reason is not obvious, this view is not, in general, correct, because relative interest-rate movements should be offset by exchange-rate movements, which are all related according to the interest-parity condition.

others.[28] A similar conclusion applies to liquidity preference. That is, to the extent that interest parity does not hold because foreign investments are less liquid, those investors for whom liquidity is not relevant can enjoy higher yields. We can see that it can pay to shop around when investing or borrowing, but it all depends on the specific circumstances of the investor or borrower.

SUMMARY

1 Forward exchange markets allow short-term investors and borrowers to avoid foreign exchange risk and exposure.

2 An investor should be indifferent with respect to investing in domestic or foreign currency when the domestic-currency interest rate equals the foreign-currency rate plus the annualized forward exchange premium or discount on the foreign currency. The investor should invest in domestic currency when the domestic-currency interest rate exceeds the foreign-currency rate plus the forward premium/discount on the foreign currency, and vice versa.

3 A borrower should borrow in foreign currency when the domestic-currency interest rate exceeds the foreign-currency rate plus the forward foreign exchange premium or discount, and borrow in domestic currency when the domestic-currency interest rate is lower than the foreign-currency interest rate plus the forward foreign-exchange premium or discount.

4 Covered interest arbitrage involves borrowing in one currency to invest in another. It is profitable when there are differences between covered borrowing costs and investment yields.

5 The covered interest-parity condition states that there will be no advantage to borrowing or lending in one currency rather than another. Forces set up by interest arbitragers will move the money and foreign exchange markets towards covered interest parity.

6 The uncovered interest-parity condition states that the differences between interest rates equals the market's expected change in the exchange rate.

7 The uncovered interest-parity condition and the PPP condition in terms of expected inflation can be used to derive the equality of real rates of return between countries. This latter relationship is the Fisher-open condition, which has an independent rationale.

8 If round-trip arbitrage were the only force moving exchange rates and interest rates towards interest parity, the deviations could be relatively large. This is because it would be necessary to be compensated in the covered interest differential for foreign exchange transaction costs and borrowing–lending spreads on interest rates.

9 One-way interest arbitrage involves choosing between alternative ways of going from current dollars/pounds to future pounds/dollars. Because the choices involve the same transaction costs whichever route is taken, one-way arbitrage should drive markets very close to covered interest parity.

10 Third-country investors or borrowers who face the same transaction costs whichever currency of denomination they choose for borrowing or investment should also drive markets very close to covered interest parity.

11 Political risks can also cause deviations from interest parity between countries, and allow a band around the interest-parity line, because investors from each country need compensation for the greater risk of investing in the other country. However, if investors outside the two countries view the countries as equally risky politically, they will drive markets to interest parity.

12 Withholding taxes do not normally affect the interest-parity condition.

13 For those who face differential taxes on income versus capital gains, the relevant interest-parity line has a different slope than the conventional interest-parity line. However, since banks pay the same tax on interest and foreign exchange gains, interest disparity should not result from differential taxes.

14 Covered foreign-currency investments are less liquid than domestic-currency investments because extra exchange transaction costs and uncertainties are met on liquidating covered foreign-currency securities. The liquidity relates to expected rather than actual transaction costs.

15 Each investor or borrower must evaluate opportunities from his or her own perspective of transaction costs, taxes, political risks, and liquidity concerns. There can be advantages from shopping around.

28 Those that can exploit differences are third-country investors for whom the political risks are similar, and borrowers for whom political risks are irrelevant. Borrowers will tend to borrow in the low-risk countries because covered interest costs there can remain lower than elsewhere.

REVIEW QUESTIONS

1 What is the "money market?"

2 How does a forward-covered investment avoid exchange rate risk?

3 According to covered interest parity, if the British pound is at a forward premium vis-à-vis the US dollar, what do we know about pound versus dollar interest rates?

4 What is "covered interest arbitrage?"

5 Does the covered interest-parity condition arise from different perspectives? Name them.

6 If the covered yield on pound-denominated securities were to become temporarily higher than on dollar securities, what would happen to interest rates and the spot and forward exchange rates?

7 What is "uncovered interest-parity?"

8 What is the "Fisher-open condition?"

9 What conditions together imply the Fisher-open condition?

10 What is "round-trip interest arbitrage?"

11 Can transaction costs affect interest parity?

12 Does political risk affect covered yields on different currency-denominated deposits at London banks?

13 Does political risk affect covered yields on US government versus Canadian government bills?
14 What is a "withholding tax credit?"
15 Do low rates of withholding tax affect covered interest parity?
16 What is "liquidity preference," and can it affect covered interest parity if people can borrow without penalty against covered foreign-currency assets?

ASSIGNMENT PROBLEMS

1 Derive the criteria for making covered money-market investment and borrowing decisions when the exchange rates are given in European terms. Derive the equivalent of equation (8.4).
2 You have been given the following information:

$r_\$$	$r_£$	$S(\$/£)$	$F_{1/4}(\$/£)$
5%	6%	1.5000	1.4985

where $r_\$$ = annual interest rate on 3-month US-dollar commercial paper
$r_£$ = annual interest on 3-month British-pound commercial paper
$S(\$/£)$ = number of dollars per pound, spot
$F_{1/4}$ ($/£) = number of dollars per pound, 3-months forward
On the basis of the precise criteria:
a In which commercial paper would you invest?
b In which currency would you borrow?
c How would you arbitrage?
d What is the profit from interest arbitrage per dollar borrowed?

3 a Use the data in Question 2 and the precise formula on the right-hand side of equation (8.4) to compute the covered yield on investment in pounds. Repeat this using the approximate formula on the right-hand side of equation (8.5).
b Compare the error between the precise formula and the approximate formula in "a" above with the error in the situation where $r_\$ = 15$ percent, $r_£ = 16$ percent, and $S(\$/£)$ and $F_{1/4}(\$/£)$ are as above.
c Should we be more careful to avoid the use of the "interest plus premium or minus discount" approximation in equation (8.5) at higher or at lower interest rates?
d If the interest rates and the forward rate in Question 2 are for 12 months, is the difference between equation (8.4) and equation (8.5) greater than when we are dealing with 3-month rates?

4 Derive the equivalent of Table 8.1 where all covered yields are against pounds rather than dollars. This will require computing appropriate cross spot and forward rates.
5 Draw a figure like Figure 8.4 to show what interest arbitrage will do to the interest-rate differentials and the forward premiums at points A to F in the table below. If all the adjustment to the interest parity occurs in the forward exchange rate, what will $F_{1/12}(\$/£)$ be after interest parity has been restored?

	A	B	C	D	E	F
$S(\$/£)$	1.6200	1.6200	1.6200	1.6200	1.6200	1.6200
$F_{1/12}(\$/£)$	1.6220	1.6150	1.6220	1.6150	1.6180	1.6120
$r_\$$(1 month), %	8.00	8.00	8.00	8.00	8.00	8.00
$r_£$(1 month), %	10.00	9.00	8.00	7.00	6.00	5.00

6 Write down the expectations form of PPP, the uncovered interest-parity condition, and the Fisher-open condition. Derive each one from the other two.
7 Assuming that there are a large number of third-country borrowers and investors, do you think that political risk will cause larger deviations from interest parity than are caused by transaction costs?
8 If banks are as happy to advance loans that are secured by domestic-currency money-market investments as they are to advance loans secured by similar foreign-currency covered money-market investments, will firms prefer domestic-currency investments on the grounds of liquidity?
9 How does the importance of liquidity relate to the probability that cash will be needed?
10 Use the framework of Figure 8.7 to show how the band within which one-way arbitrage is unprofitable compares to the band within which round-trip arbitrage is unprofitable.
11 Why might a borrower want to borrow in a currency that is at a forward discount if that borrower faces a higher tax rate on interest income than on capital gains?
12 Why does the Fisher-open condition relate to countries rather than currencies?
13 In general, are transactions costs higher in spot or forward markets? Does this hold any implications for whether interest parity will hold exactly?
14 What role does the rest of the world play in determining whether covered interest parity will hold between any two currency-denominated securities?
15 Suppose that real interest rates are equal for all countries in the world. Does this imply anything for the relationship between covered interest-rate parity and the PPP condition?

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Part IV

Managing foreign exchange risk and exposure

Until this point our concern has been with the nature of markets in which currencies and currency derivatives trade. The focus of the remainder of the book is with managerial issues, such as using international financial markets to deal with the special opportunities and risks of international trade and investment. We shall see on numerous occasions on this journey through international managerial finance that an understanding of the financial markets and environment are essential elements of financial decision making. The chapters in Part IV begin our journey with a discussion of the objectives of international financial management and how to achieve these objectives. The focus is on operating issues, from the measurement of foreign exchange risk and exposure, to the methods and potential for success of currency speculation.

The opening chapter of Part IV, Chapter 9, takes us directly to the meaning of foreign exchange exposure. It is shown that exposure is a measure of the sensitivity of changes in domestic currency values of assets and liabilities to unanticipated changes in exchange rates. We show that surprisingly, domestic as well as foreign financial instruments can face foreign exchange exposure, and that under special circumstances, foreign financial instruments may *not* be exposed. The chapter also explains foreign exchange risk, a matter which is often confused with exposure. Exchange-rate risk is shown to relate to the *variability* of domestic currency values of assets and liabilities, whereas exposure is the *amount at risk*. This makes risk and exposure conceptually and even dimensionally different: the two concepts have different units of measurement.

Chapter 10 provides a brief introduction to international accounting principles and how these impact on exposure as reflected in financial accounts – accounting exposure – versus true, underlying value changes – real exposure. While it is real exposure that should really matter to the owners of a multinational firm, values that show up in accounting statements cannot be ignored when, for example, taxes are based on operating incomes as expressed in a company's "reporting currency."

Chapter 11 is devoted to the effect of exchange rates on sales and operating profitability, so-called "operating exposure." Use is made of the microeconomic theory of the firm which emphasises marginal cost and marginal revenue. It is shown that the amount of operating exposure depends on such factors as elasticity of demand and flexibility of production. The chapter ends with a consideration of situations where exchange rates have different effects on firms in the short-run versus the long-run. This short-run versus long-run distinction for firms is analogous to the short-run versus long-run distinction for the economy as a whole, met in Chapter 6 when we explained the J curve.

With exposure and risk defined and explained in Chapters 9, 10, and 11, in Chapter 12 we shift attention to the management of exposure and risk. We begin by asking whether managers should hedge